Clasps/fasteners (snaps, zippers, buttons) used in infant products: HMTc Standard

Karen Pendergrass · · Heavy Metal Tested & Certified, Institute of Contaminant Standards

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for clasps/fasteners (snaps, zippers, buttons) used in infant products in Infant and Child Durable Goods and Textiles (Ages 0-5), evaluated on total element content of the fastener component and its plating or coating (mg/kg of component; dried non-volatile film for platings and lacquers): a content basis, not an extraction, nickel-release or migration basis. 5 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only): the strictest applicable maximum set by a credible sovereign government for this exact product form in the row's declared native basis; a disclosed read-across from the nearest applicable sovereign cell where the exact cell is unregulated; a substantiated not-material determination where the analyte is a species bounded by a published parent; and a control with a named reference method where neither a sovereign value nor a supportable read-across exists. Never occurrence-derived.

2 | Master limit table

All values in ppb (µg/kg), as_sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	90,000	government maximum level

Cadmium (Cd)	Tier 1	75,000	human-grade read-across ()
Total arsenic (tAs)	Tier 2	25,000	human-grade read-across (AU)
Inorganic arsenic (iAs)	Tier 1	≤ tAs	bounded by tAs (reflex speciation)
Mercury, total (tHg)	Tier 1	10,000	human-grade read-across (Canada)
Methylmercury (MeHg)	Tier 1	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	Tier 2	930,000	human-grade read-across (European Union)
Tin (Sn)	Tier 2	180,000,000	human-grade read-across (European Union)
Aluminium (Al)	Tier 2	28,130,000	human-grade read-across (European Union)
Total chromium (Cr)	Tier 2	60,000	screening ceiling (AU)
Hexavalent chromium (Cr-VI)	Tier 2	3,000	human-grade read-across (European Union)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	90,000	government maximum level
Cadmium (Cd)	75,000	— human-food analog
Total arsenic (tAs)	25,000	AU — human-food analog
Inorganic arsenic (iAs)	≤ tAs	iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell.
Mercury, total (tHg)	10,000	Canada — human-food analog

Methylmercury (MeHg)	≤ tHg	MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell.
Nickel (Ni)	930,000	European Union — human-food analog
Tin (Sn)	180,000,000	European Union — human-food analog
Aluminium (Al)	28,130,000	European Union — human-food analog
Total chromium (Cr)	60,000	government maximum level
Hexavalent chromium (Cr-VI)	3,000	European Union — human-food analog

4 | Cite this standard

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References

[1] AU — human-food maximum levels
Binds tAs. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] Canada — human-food maximum levels
Binds tHg. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] European Union — human-food maximum levels
Binds Ni, Sn, Al, Cr-VI. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

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COMPETING INTERESTS

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DATA AVAILABILITY

PUBLICATION HISTORY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/clasps--fasteners-infant-products. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/-clasps-fasteners-infant-products.